

Technical Supplement: Synthetic Pointwise Topology Selection and Identifiability Limits

Solar-Planetary Phase-Partition Theory with ASTRA

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Status statement. Every result in this supplement is synthetic. No Solar System, exoplanet, laboratory, or mission data are fitted. The tests demonstrate implementation behavior under declared assumptions; they do not validate a planetary topology.

1 Purpose and relation to the main paper

The main paper proposes **Solar-Planetary Phase-Partition Theory (SPPT)** and names its inference and validation layer **ASTRA — Astronomical State-Topology and Reservoir Analysis**. SPPT supplies the physical state representation, conservation rules, thermodynamic admissibility conditions, and topology-transition logic. ASTRA asks whether candidate phase-reservoir graphs can be distinguished from boundary observations and whether their added structure produces calibrated predictive improvement.

This supplement documents three narrow implementation tests:

1. a transparent three-reservoir pointwise topology-selection benchmark with training-BIC selection and a distinct post-selection held-out forcing;
2. a 64-seed Monte Carlo robustness repetition of that benchmark;
3. a two-reservoir frequency-domain demonstration of static and single-frequency-amplitude non-identifiability.

The tests are deliberately small enough to audit line by line. They are not a substitute for equations of state, phase diagrams, atmosphere-interior evolution, radiative transfer, or mission-data likelihoods.

2 Three-reservoir pointwise topology-selection benchmark

2.1 Candidate node-labeled support hypotheses

The benchmark contains three reservoirs, indexed from deep to observable surface:

$$0 = \text{deep}, \quad 1 = \text{intermediate}, \quad 2 = \text{surface}.$$

Four connected, node-labeled undirected transport-support hypotheses are compared. The node labels and capacities are part of each hypothesis: as abstract unweighted graphs, the chain and deep star are both a three-node path rooted at a leaf.

Candidate node-labeled graph families and their free-conductance counts.

Graph family	Edges	Free conductances
Chain	$(0, 1), (1, 2)$	2
Surface star	$(0, 2), (1, 2)$	2
Deep star	$(0, 1), (0, 2)$	2
Triangle	$(0, 1), (1, 2), (0, 2)$	3

The generating graph is the chain, with normalized conductances

$$k_{\text{true}} = (0.22, 1.40).$$

Node capacities, surface loss, internal power, and observational noise are

$$C = (8, 3, 1), \quad \lambda_s = 1.2, \quad P_d = 1.0, \quad \sigma_y = 2.5 \times 10^{-3}.$$

2.2 Dynamics

For graph Laplacian $L_{\mathcal{G}}(k)$ and surface-loss matrix

$$A = \text{diag}(0, 0, \lambda_s),$$

the state satisfies

$$\dot{CT} = -(L_{\mathcal{G}} + \Lambda)T + \begin{bmatrix} P_d \\ 0 \\ f(t) \end{bmatrix}.$$

Only the surface state $T_2(t)$ is observed. The training and held-out forcings are intentionally different:

$$\begin{aligned} f_{\text{train}}(t) &= 0.35 \sin(0.55t) + 0.18 \mathbf{1}_{t > 10} - 0.12 \mathbf{1}_{t > 23}, \\ f_{\text{test}}(t) &= 0.28 \sin(1.05t) + 0.22 \sin(0.19t) + 0.20 \mathbf{1}_{6 < t < 14}. \end{aligned}$$

The training window contains 361 samples over $0 \leq t \leq 36$; the held-out window contains 261 samples over $0 \leq t \leq 26$.

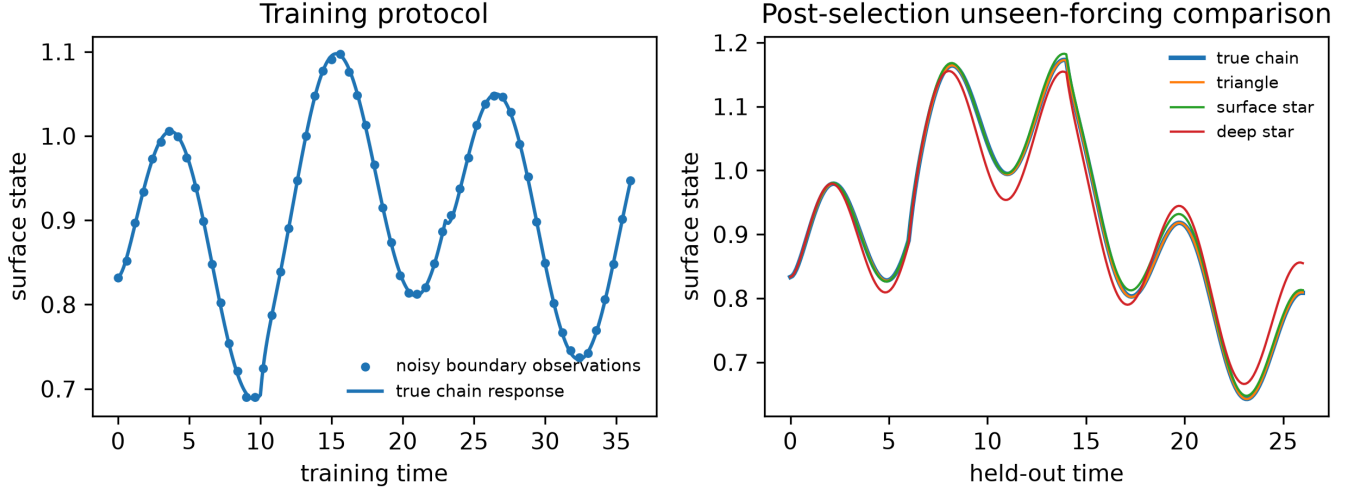
2.3 Fitting and selection

Positive conductances are parameterized as $k_e = \exp \eta_e$ and fitted by nonlinear least squares using only the noisy training surface series. All generation constants, seeds, candidates, and evaluation code are public; the benchmark is not blinded or external validation. Each fit uses a release-frozen 20-start generic design combining low-discrepancy coverage with unit and coordinate-wise decade anchors, together with exact forward sensitivities of the implemented propagator to log conductance. The 20-start design was adopted during release audit after replay of this same synthetic benchmark exposed a missed endpoint under the earlier 12-start design. The added anchors were therefore informed by benchmark behavior, and the reported reruns are regression evidence for the repaired implementation rather than untouched evaluation. Write the scalar solver cost as J (SciPy's least-squares cost, equal here to one half of the residual sum of squares) and the solver-reported first-order optimality as O_{SciPy} (for the bounded trust-region-reflective method, SciPy reports the infinity norm of its bound-aware scaled gradient). A solver result is eligible only when it reports positive termination status, finite parameters and objective, and $O_{\text{SciPy}} / \max(1, J) \leq 10^{-4}$; the lowest-cost eligible start is retained. If a non-eligible endpoint has a cost lower by more than 10^{-4} of the retained cost scale, the fit fails closed as insufficient optimizer coverage. Every start vector, solver disposition, endpoint, cost, optimality diagnostic, and active-bound mask is retained in both duplicate machine-readable serializations. With residual sum of squares RSS, sample count n , and free-conductance count p , the benchmark reports

$$\text{BIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + p \ln n.$$

This BIC is a small-sample implementation choice, not a universal ASTRA selection law. Its regular-model interpretation is only approximate here because the triangle nests the chain at a conductance boundary, candidate families intersect exactly, and pole-zero-cancellation strata are singular. At an exact input-output equivalence, BIC can select a smaller representation but cannot establish which physical support generated the data. The main framework requires structural-identifiability checks, posterior predictive calibration, and physically matched controls for real inference.

Synthetic pointwise topology-selection benchmark



Training observations and held-out predictions for the four candidate node-labeled support hypotheses. The left panel compares noisy surface observations with the true-chain training response; the right panel compares all four candidate predictions with the true chain under unseen forcing. The benchmark is synthetic and uses only the surface node as an observation.

The complete plotted benchmark series and fit diagnostics are provided in `data/synthetic_topology_benchmark.csv` and `data/synthetic_topology_benchmark.json`.

2.4 Exact static degeneracy

At zero external forcing, every connected candidate graph transports the same total internal power to the same surface sink. Consequently, all four have the same surface equilibrium:

$$T_{s,\text{eq}} = \frac{P_d}{\lambda_s} = \frac{1}{1.2} = 0.833333\dots$$

Their hidden deep states are not the same.

For this static comparison only, the code uses the declared true conductances for the chain and assigns conductance 0.7 to every edge of each non-generating candidate. These illustrative alternative weights do not enter the fitted benchmark results.

Static surface and deep equilibria under the declared comparison conductances.

Graph family	Surface equilibrium	Deep equilibrium
Chain	0.833333	6.093074
Surface star	0.833333	2.261905
Deep star	0.833333	2.261905
Triangle	0.833333	1.785714

This is the benchmark's core inverse-problem fact: a static boundary value can identify total throughput without identifying the internal transport architecture.

2.5 Algebraic-statistical audit of dynamic identifiability across supports

Algebraic statistics treats each candidate family as a parameter space mapped into observable coordinates and asks about the fibers of that map. A singleton admissible fiber gives global identifiability; a finite fiber with more than one point gives discrete ambiguity; a positive-dimensional fiber gives continuous non-identifiability; and changes in fiber dimension mark singular strata where regular likelihood asymptotics can fail. This global fiber question is stronger than checking a local Jacobian rank or observability matrix at one realization.

Observability inside one fixed state-space realization is therefore not the same as structural identifiability across parameterized graph realizations. The algebraic-statistical distinction can be resolved exactly for this three-node model. Write

$$a = k_{01}, \quad b = k_{12}, \quad c = k_{02},$$

and let node 2 be both the forced and observed surface port. After subtracting each graph's equilibrium, the complete surface input-output law is the rational transfer function

$$H(s) = e_2^T [s \operatorname{diag}(C_0, C_1, C_2) + L + \lambda_s e_2 e_2^T]^{-1} e_2.$$

Define

$$\alpha = (C_0 + C_1)a + C_0b + C_1c, \quad \tau = ab + ac + bc, \quad d = b + c.$$

A direct cofactor expansion gives

$$H(s) = \frac{C_0 C_1 s^2 + \alpha s + \tau}{C_0 C_1 C_2 s^3 + [C_2 \alpha + C_0 C_1 (d + \lambda_s)] s^2 + [(C_0 + C_1 + C_2) \tau + \lambda_s \alpha] s + \lambda_s \tau}.$$

The single surface port therefore determines only (α, τ, d) before possible pole-zero cancellation, not necessarily the physical edge conductances. With the benchmark capacities, two different two-edge hypotheses give the same exact triple:

$$\begin{aligned}
\text{surface star:} & \quad (k_{02}, k_{12}) = (5, 6), \\
\text{deep star:} & \quad (k_{01}, k_{02}) = (30/11, 11), \\
(\alpha, \tau, d) = & \quad (63, 30, 11).
\end{aligned}$$

Both consequently have

$$H(s) = \frac{24s^2 + 63s + 30}{24s^3 + (1779/5)s^2 + (2178/5)s + 36}.$$

They are connected, have equal parameter count, and are minimal third-order realizations. Their hidden equilibria and trajectories differ, but their surface trajectories are identical for every surface forcing when each run begins at its own equilibrium. Thus changing the training waveform, adding noiseless samples, or applying a BIC penalty cannot distinguish this pair.

The ambiguity is not an isolated coincidence. For a triangle, the second algebraic branch preserving (α, τ, d) is

$$b^* = \frac{2C_1 d}{C_0 + C_1} - b, \quad c^* = d - b^*, \quad a^* = a + \frac{C_0 - C_1}{C_0 + C_1}(b - b^*),$$

whenever the transformed conductances remain nonnegative. For example, the two strictly positive triangles

$$(a, b, c) = (1, 1, 1), \quad (a^*, b^*, c^*) = (171/121, 1/11, 21/11)$$

have the same surface transfer. Each point is locally full-rank, yet the global inverse is two-to-one. On the singular balance locus $C_0 b = C_1 c$ —here $8k_{12} = 3k_{02}$ —an internal mode cancels from the port response and k_{01} becomes completely invisible. A local Fisher-information or observability-rank check alone therefore cannot rule out global cross-realization ambiguity.

The released generating chain remains a fair pointwise test. Its $(a, b, c) = (0.22, 1.40, 0)$ has algebraic partner $b^* = -7/11$, outside the nonnegative-conductance domain. The frozen 64/64 result is therefore preserved as selection of this distinguishable generating point within the declared candidate set; it is not evidence that the candidate families are globally topology-identifying. A spatially distinct time-varying input, another observation channel, an intervention, or independently justified structural constraints may separate the displayed equivalences, but structural identifiability must be recomputed for the particular augmented design.

2.6 Single-seed pointwise selection

The released realization uses seed 20260801. Results ranked by BIC are:

Single-seed synthetic candidates ranked by training-set BIC; held-out RMSE is reported only as a post-selection comparison.

Rank	Graph	Fitted conductances	Training RMSE	BIC	Held-out RMSE
1	Chain	0.228876; 1.393863	0.002500	-4314.159	0.000474
2	Triangle	0.225037; 1.390020; 0.001869	0.002499	-4308.513	0.000452
3	Surface star	0.087328; 1.196997	0.005059	-3805.156	0.006720
4	Deep star	0.045813; 1.084790	0.019802	-2819.876	0.025147

The overconnected triangle has a slightly smaller held-out RMSE than the chain in this one noise realization. That does **not** identify the triangle. Its extra shortcut is fitted at 0.001869, approximately 0.85% of the weaker true chain edge, and the training-BIC penalty selects the two-edge chain. The correct interpretation is minimum-representation selection at the released generating point, with an overconnected control that collapses toward that point.

3 Monte Carlo robustness ensemble

The complete fit was repeated for 64 consecutive independent Gaussian-noise seeds, beginning at 20260801 and preserving all physical and numerical settings. Selection uses training BIC only; the noiseless held-out generating response is compared afterward. The chain was the minimum-BIC graph in 64 of 64 realizations.

The median separation between the overconnected and minimum graph was

$$\text{median}(\text{BIC}_{\text{triangle}} - \text{BIC}_{\text{chain}}) = 5.8493.$$

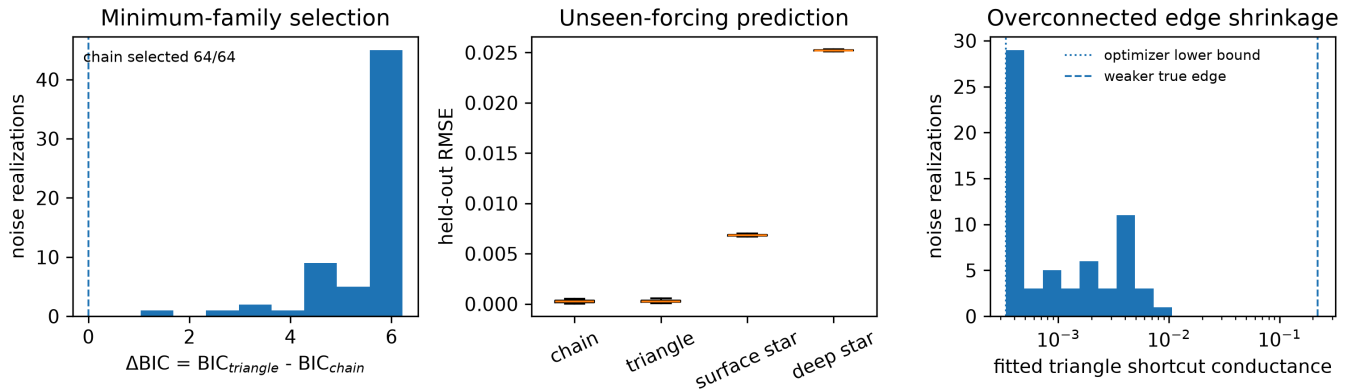
The median triangle shortcut conductance was

$$7.6485 \times 10^{-4},$$

showing systematic shrinkage of the unnecessary edge toward zero under the released optimization and noise scale. The shortcut reaches the declared lower bound $\exp(-8)$ in 29 of 64 realizations, so the distribution is censored. The full output preserves these bound-active fits and does not treat them as interior measurements.

The post-selection unseen-forcing comparison preserves a material negative outcome: the triangle has a smaller held-out RMSE than the chain in 23 of 64 realizations. Mean held-out RMSE is 2.4864×10^{-4} for the chain and 2.6969×10^{-4} for the triangle. The observed ΔBIC range is 1.0342 to 6.2157. These facts do not alter the training-BIC winner, but they limit what 64/64 selection establishes.

Synthetic pointwise topology-selection ensemble



Across 64 synthetic noise realizations, BIC selects the minimum chain in every run. The left panel shows the positive triangle-minus-chain BIC differences, the center panel compares held-out error distributions for all candidates, and the right panel shows the triangle shortcut shrinking toward or reaching the optimizer's lower bound.

Per-seed selections, errors, shortcut conductances, and optimizer diagnostics are provided in `data/synthetic_topology_ensemble.csv` and `data/synthetic_topology_ensemble.json`.

This ensemble does not estimate a general false-positive rate. It holds capacities, sink location, forcing, noise law, and the candidate graph set fixed. The CSV and JSON are alternate serializations of the same 64 runs, not independent evidence. A real ASTRA validation program must vary those assumptions, include misspecified models, and test graph-posterior calibration.

4 Frequency-domain parameter-localization demonstration

4.1 Linear response

For a fixed graph, capacities C , transport Laplacian L_K , and local-loss matrix Λ , consider

$$\dot{CT} = P(t) - (L_K + \Lambda)T.$$

With harmonic forcing $P(t) = \Re(b e^{i\omega t})$ and scalar observation $y = c^\top T$, the transfer response is

$$H(i\omega) = c^\top (i\omega C + L_K + \Lambda)^{-1} b.$$

Amplitude and phase depend on the eigenmodes, capacities, and connectivity. Static gain corresponds only to $\omega = 0$ and can leave deep parameters unconstrained.

4.2 Two-reservoir demonstration

The normalized generating model uses

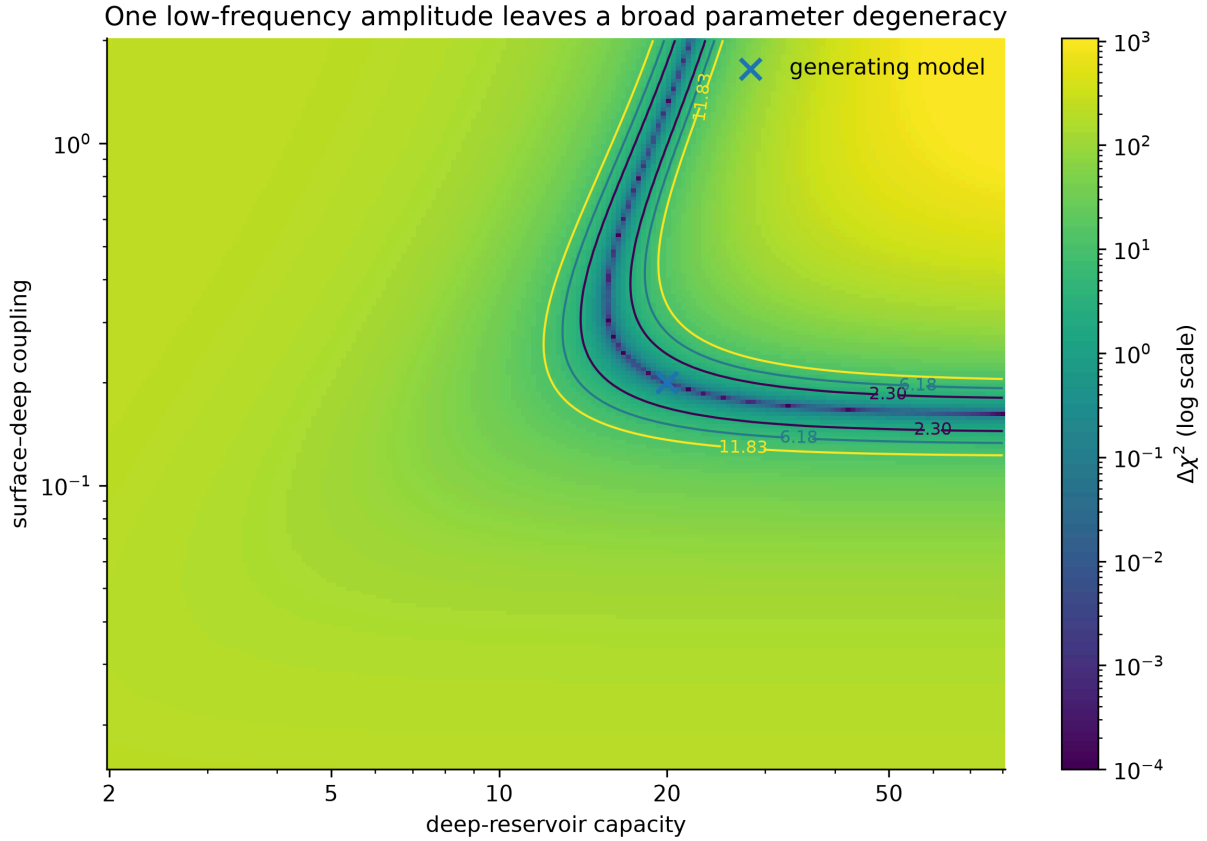
$$C_s = 1, \quad C_d = 20, \quad k = 0.2, \quad \lambda_s = 1.$$

Its two decay rates and relaxation times are:

Decay modes of the normalized two-reservoir generating model.

Mode	Decay rate	Relaxation time
Slow	0.0083217	120.1678
Fast	1.2016783	0.83217

A single low-frequency amplitude at $f = 0.003$ leaves a broad valley in (C_d, k) space. The best point on the declared grid is $(16.0319, 0.271916)$, visibly displaced from the generating pair despite matching that one response feature.

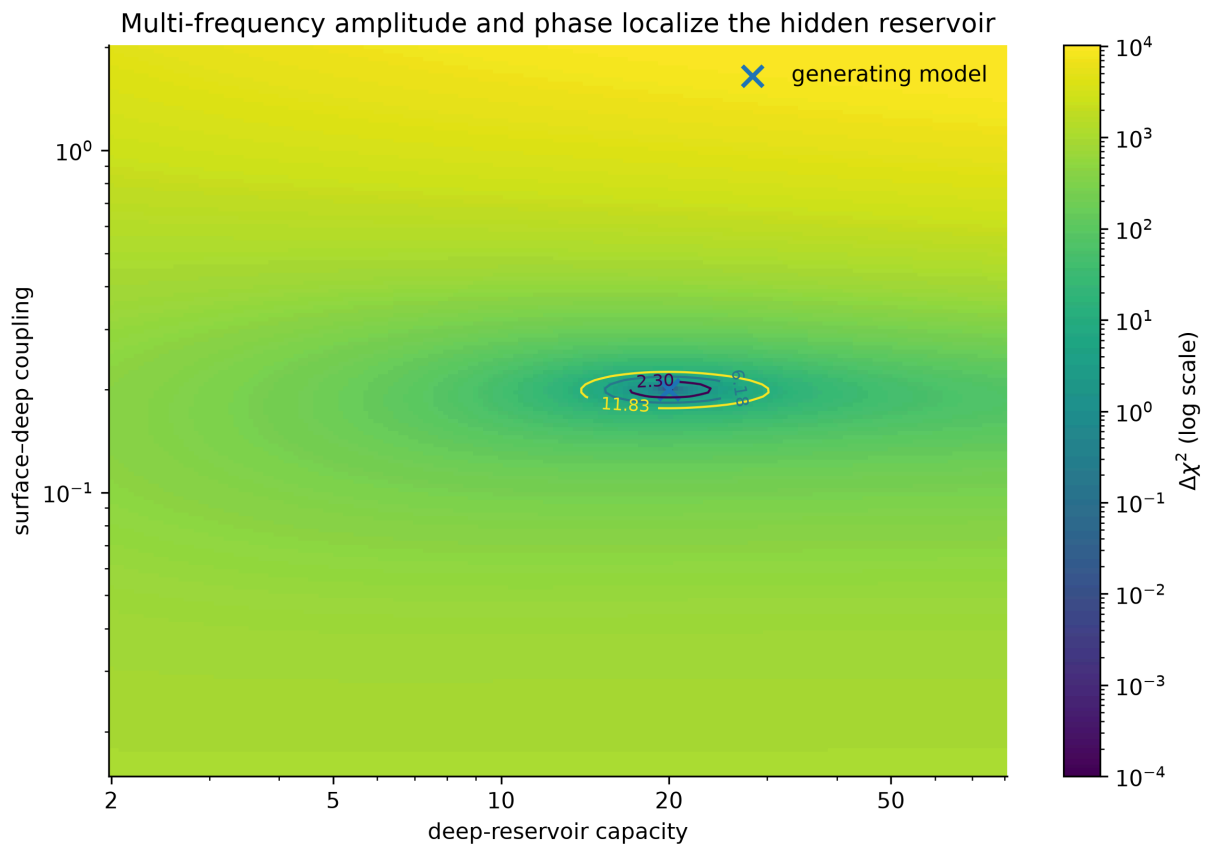


One low-frequency amplitude leaves a broad capacity-coupling degeneracy. Color represents the objective increase on a logarithmic scale, labeled contour lines mark fixed objective increases, and a cross marks the generating parameters.

Using complex amplitude and phase at 24 logarithmically spaced frequencies localizes the grid minimum near the generating model:

$$(C_d, k)_{\text{grid best}} = (20.1111, 0.201304).$$

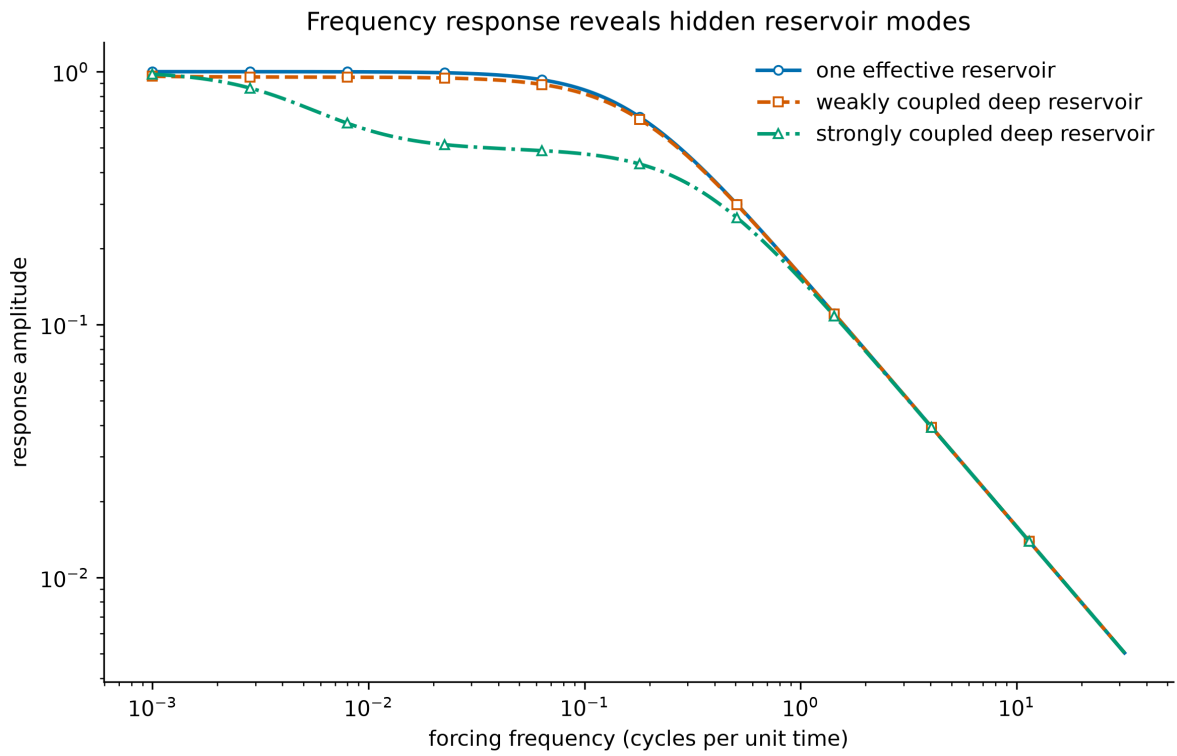
The one-frequency objective deliberately discards phase and retains only amplitude. This comparison does not show that multiple frequencies are structurally necessary: one exact nonzero complex response supplies two real features and can generically constrain the two unknown parameters in this normalized model. Multiple frequencies provide redundancy, conditioning, and noise resistance; the released figures demonstrate finite-grid localization, not a minimal-data theorem.



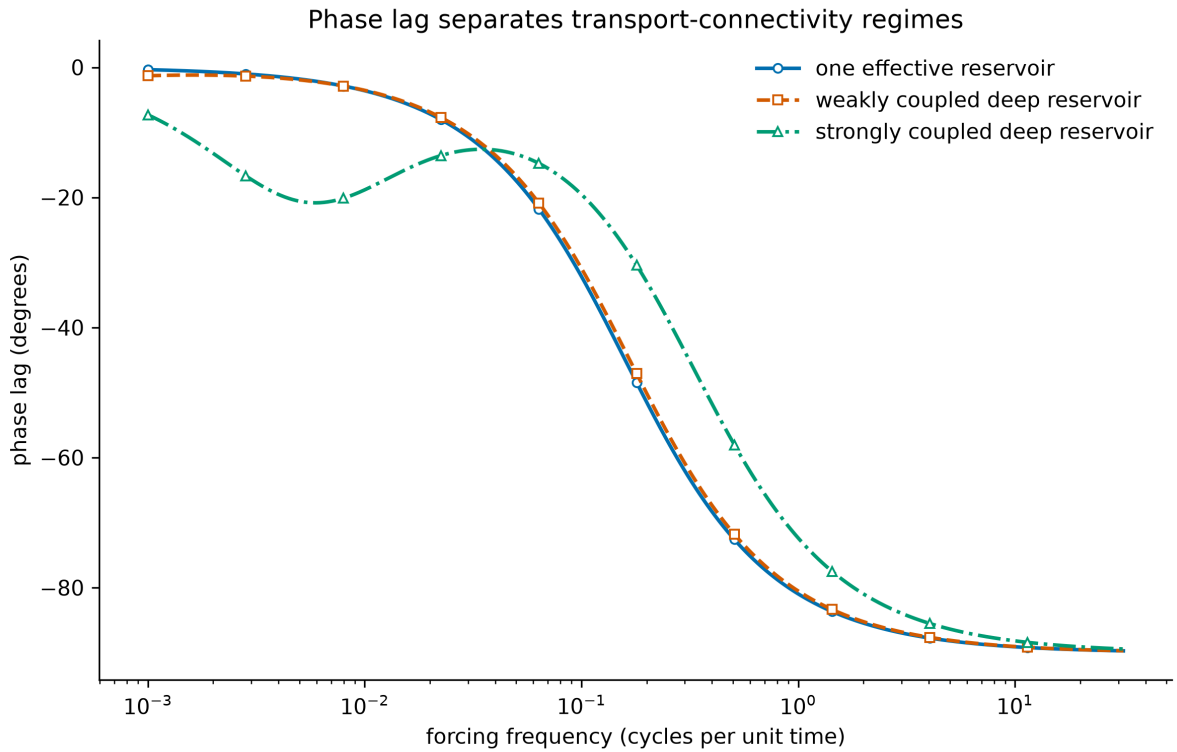
Multi-frequency amplitude and phase localize the generating two-reservoir parameters in the synthetic grid search. Color represents the objective increase on a logarithmic scale, labeled contours close around a compact minimum, and a cross marks the generating parameters.

The objective value at every capacity-conductance grid point for both demonstrations is provided in `data/inverse_objective_landscapes.csv`; the generating and best-grid parameters are summarized in `data/inverse_demo_summary.json`.

The mechanism is visible directly in the response curves. Adding a weakly or strongly coupled deep reservoir changes both attenuation and phase lag over a range of forcing frequencies.



Amplitude response for one-reservoir and two-reservoir normalized models. Solid lines with circles identify the one-reservoir model, dashed lines with squares the weakly coupled deep reservoir, and dash-dot lines with triangles the strongly coupled deep reservoir.



Phase response for the same three normalized models, using the same solid-circle, dashed-square, and dash-dot-triangle mapping as the amplitude plot.

Forcing frequency, model name, amplitude, and phase in radians are provided in `data/frequency_response.csv`.

The demonstration does not imply that astronomical observations can freely choose forcing frequencies. In practice, usable variation may come from orbital phase, seasons, eclipses, stellar variability, secular cooling, impacts, atmospheric events, or comparisons across a population. The input spectrum and observation operator must be modeled rather than assumed.

5 Numerical implementation validation

The single-run benchmark uses `scipy.integrate.solve_ivp` with maximum step 0.05, relative tolerance 10^{-9} , and absolute tolerance 10^{-11} . The 64-seed ensemble uses exact matrix-exponential propagation under a four-substep zero-order-hold approximation to the time-dependent forcing. Every fitted family is evaluated from 20 distinct, fixed, graph-independent multistarts combining low-discrepancy coverage with unit and coordinate-wise decade anchors. A retained fit is rejected if a non-admitted endpoint produces a materially lower cost. As disclosed above, this strengthened start design and its reruns are post-audit regression evidence. Release generation uses CPython 3.12.10, Git for Windows 2.55.0.windows.3, and probed single-thread NumPy and SciPy OpenBLAS Haswell kernels on compatible Windows x86-64 hardware; the byte-identity claim is limited to the exercised release outputs under that complete frozen runtime.

The fast propagator was compared with the high-accuracy solver at the generating parameters:

Fast-propagator errors relative to the high-accuracy solver.

Protocol	Maximum surface error	Surface RMSE	Maximum error / noise SD
Training forcing	9.63×10^{-6}	6.70×10^{-6}	0.00385
Held-out forcing	1.60×10^{-5}	9.70×10^{-6}	0.00640

Thus the numerical approximation contributes less than 0.7% of the observational-noise standard deviation in the validation cases.

6 Automated tests

The release's complete discovered test suite covers:

- conservation of a declared weighted inventory under internal transport and balanced reaction;
- the exact periodic-trap solution and loop-area integral;
- the weak-cut spectral bound;
- ideal electroreduction scale calculations;
- static boundary degeneracy;
- existence of a negative differential transport region in the declared toy closure;
- graph-Laplacian conservation and positive semidefiniteness;
- two-reservoir steady state, step response, poles, and frequency response;
- Fisher-information symmetry and positive semidefiniteness;
- fast-propagator agreement with the high-accuracy benchmark;
- minimum-chain selection in the released seed.
- the complete state-dependent derivative, including $K(1 - dT_u/dT_d)$;
- frozen multistart design, convergence and materially-better-endpoint rejection, active-bound reporting, numeric-kernel override and probe checks, and negative release-integrity checks.

Passing tests establish consistency with the implemented equations. They do not establish that the equations are sufficient for a specific planet.

7 Reproducibility commands

From the release root, install the exact hash-locked dependencies and run the canonical verification:

```
python -m pip install --require-hashes -r requirements-lock.txt
python -I -B tools/verify.py --all
```

For a direct scientific replay, `python scripts/make_figures.py` regenerates every figure and benchmark output; the wrapper declares the repository import path and frozen build environment itself. The scripts write data to `data/` and figures to `figures/`. Random seeds, all normalized parameter values, accepted-start counts, first-order optimality values, and active-bound flags are stored in machine-readable outputs.

8 Limitations and required extensions

The benchmark is intentionally favorable. Its main limitations are:

1. **Known capacities and sink structure.** Only conductances and graph family are fitted.
2. **Closed candidate set.** The generating graph is present among four candidates.
3. **Linear dynamics.** No state-dependent phase boundary, topology guard, or nonlinear radiation is included.
4. **Single collocated boundary channel.** Only the surface state is forced and observed; its noise is independent Gaussian noise with known scale.
5. **Structural non-identifiability.** Even with known capacities and perfect surface data, exact positive cross-support equivalences and a singular dark-mode locus remain. The released chain is a distinguishable point, not a proof of family-wide topology recovery.
6. **Correct forcing model.** Input timing and amplitude are treated as known.
7. **BIC approximation.** Full Bayesian graph evidence and posterior calibration are not computed; ordinary BIC does not resolve exact equivalence classes or supply regular asymptotics on singular strata.
8. **Boundary-nested control.** The triangle collapses to the chain at a conductance boundary, so regular BIC asymptotics need not hold exactly.
9. **No physical equations of state.** The variables are normalized and not mapped to a planet.
10. **No discrepancy process.** Structural model error is absent.
11. **Limited adversarial negative controls.** Exact algebraic equivalences are now tested, but random graph families, equal-parameter nonlinear fixed graphs, and correlated-noise controls remain future work.
12. **No population validation.** The tests do not establish transfer to exoplanet or Solar System inference.

The next computational milestone is a blinded benchmark suite in which capacities, sink placement, forcing spectrum, phase thresholds, observation and intervention ports, and noise covariance vary; the candidate set sometimes omits the generating graph; exact input-output equivalence classes are reported rather than tie-broken; and selection is judged by graph-posterior calibration as well as prediction.

9 Artifact map

Public artifact map for reproduction and audit.

Artifact	Purpose
<code>src/sppt_core.py</code>	Conservation, trap, bottleneck, electroreduction, and reduced transport calculations
<code>src/astra_reservoir.py</code>	Linear ASTRA network, transient, frequency-response, and information tools
<code>src/astra_optimization.py</code>	Frozen multistart design and explicit optimizer-convergence gate
<code>scripts/synthetic_topology_benchmark.py</code>	High-accuracy single-run graph benchmark
<code>scripts/benchmark_ensemble.py</code>	64-seed robustness ensemble
<code>scripts/generate_astra_figures.py</code>	Frequency-domain and response demonstrations
<code>tests/</code>	Automated consistency and numerical-validation checks
<code>data/*.json</code> and <code>data/*.csv</code>	Machine-readable inputs and outputs
<code>figures/supplement_figure_S*.png</code>	Supplementary visualizations

10 Conclusion

The released synthetic work establishes three limited points. Static surface equilibrium can conceal substantially different internal states; transient forcing separates the released chain point from the declared alternatives but cannot globally identify these support families from the single surface port; and multi-frequency complex response can improve localization relative to one amplitude-only response feature. Within the declared benchmark, training BIC consistently selects the released minimum generating representation and an unnecessary edge shrinks toward or reaches its lower bound. Held-out results remain a post-selection comparison and preserve cases favoring the overconnected control.

Those results justify proceeding to harder blinded tests. They do not yet justify an astronomical claim. The scientific threshold remains the same as in the main paper: topology must pay predictive rent under realistic physics, uncertainty, and unseen data.